

| Question | Marking Guidance                                                                        | Mark                       | Comments                                                                                                                                                |
|----------|-----------------------------------------------------------------------------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 05.1     | <p>RNA/rRNA/ribosomal RNA</p> <p><b>AND</b></p> <p>Protein/polypeptide/amino acids;</p> | <p>1<br/>(1 x<br/>AO1)</p> | <p>Both molecules needed for <b>one</b> mark</p> <p>Accept in either order</p> <p>Reject other types of RNA e.g. mRNA or tRNA</p> <p>Reject histone</p> |

| Question | Marking Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Mark                  | Comments                                                                                                                                                                                                          |
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| 05.2     | <p>Similarities</p> <p>1. <u>DNA</u> replication;</p> <p>2. Division of the cytoplasm</p> <p><b>OR</b></p> <p>Formation of the cell-surface membrane</p> <p><b>OR</b></p> <p>One division into two cells;</p> <p>3. Synthesis of proteins/ribosomes;</p> <p>Differences</p> <p>4. No histones synthesised in binary fission</p> <p><b>OR</b></p> <p>No histones associated with DNA in binary fission;</p> <p>5. No mitosis/interphase in binary fission;</p> <p>6. No membrane-bound organelles replicated in binary fission;</p> | 4 max<br>(4 x<br>AO2) | <p>2. Allow cytokinesis</p> <p>4./5./6. Allow 'only'</p> <p>4./5./6. Accept prokaryote/bacteria for binary fission</p> <p>5. Accept a named stage of mitosis</p> <p>6. Accept named membrane bound organelles</p> |

| Question    | Marking Guidance                                                                                                                                                                                               | Mark              | Comments                                                                                                                                                                                       |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>05.3</b> | 1.(DNA is checked for) mutations;<br><br>2. (Mutations) could result in non-functioning proteins/enzymes<br><br><b>OR</b><br><br>(Mutations) could lead to uncontrolled cell division/tumours/cancerous cells; | 2<br>(2 x<br>AO2) | 1. Accept change in base sequence<br>1. Accept correctly named mutation<br>2. Accept descriptions eg change in secondary/tertiary/quaternary structure/length of polypeptide/different protein |